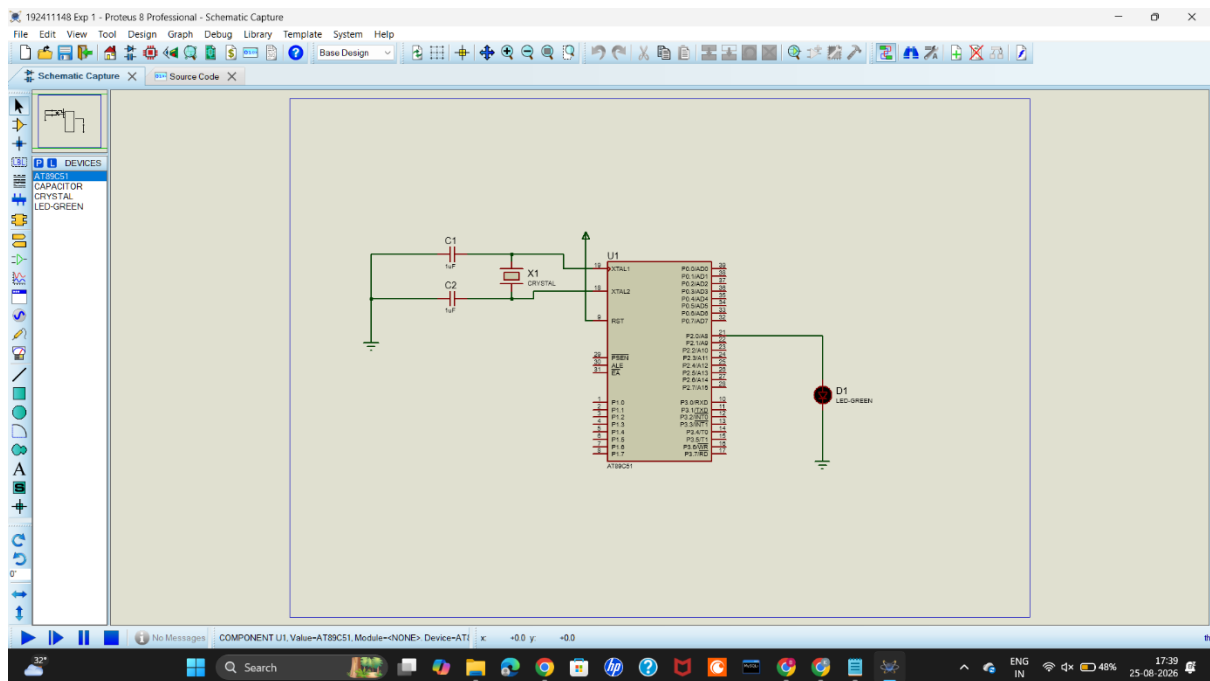


Experiment 1 Output



192411148 Exp 1 - Proteus 8 Professional - Source Code

File Project Build Edit Debug System Help

Schematic Capture Source Code

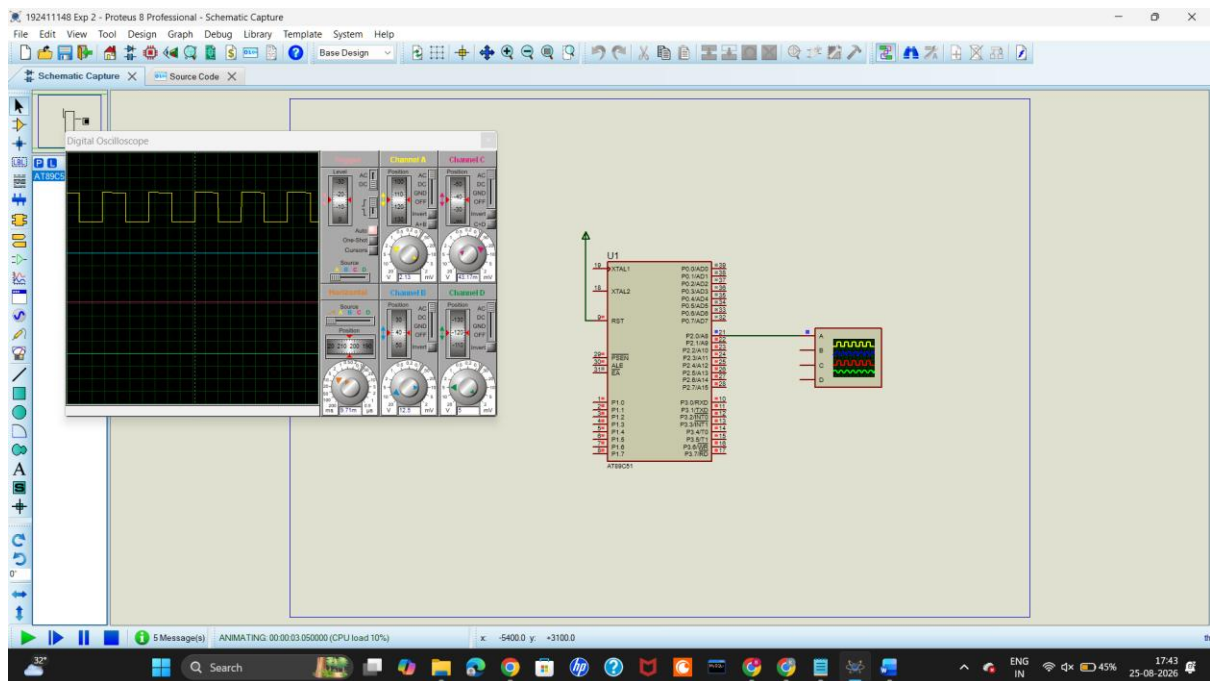
main.asm

```
1 ;Name: K Hemant Kumar
2 ;Reg No: 192411148
3 ;Code: EGA1407
4
5 ORG 0000H
6 UP: SETB P2.0
7 ACALL DELAY
8 CLR P2.0
9 ACALL DELAY
10 SJMP UP
11 DELAY: MOV R4, #35
12 H1: MOV R3, R25
13 H2: DJNZ R3, H2
14 DJNZ R4, H1
15 RET
16 END
```

VSM Studio Output:

File C:\Users\KHEMANH-1\AppData\Local\Temp\VSM Studio\986175c6e77d4eebb6238d3d65121b\AT89C51\Debug\Debug HEX not found

Experiment 2 Output



The screenshot displays the Proteus 8 Professional Source Code window for the same project. The source code is written in assembly language for the AT89C51 microcontroller. The code is as follows:

```
main.asm
1 Name: K Hemant Kumar
2 Reg No: 192411148
3 Code: EDA1407
4 ORG 0000H
5 UP: SETB P2.0
6 ACALL DELAY
7 CLR P2.0
8 ACALL DELAY
9 JMP UP
10 DELAY: MOV R4, #35
11 H1: MOV R3, #255
12 H2: DJNZ R3, H2
13 DJNZ R4, H1
14 RET
15 END
```

The VSM Studio Output window at the bottom shows the compilation results:

```
VSM Studio Output
no errors
ASEM00X.E\I: Debug list
Processed 57 lines
Compiled successfully
```

Experiment 3 Output:

192411148 Exp 3 - Proteus 8 Professional - Schematic Capture

File Edit View Tool Design Graph Debug Library Template System Help

Schematic Capture Source Code

DEVICES

AT89C51
MOTOR STEPPER
ULN2803

U1

XTAL1
XTAL2
RST
PSEN
ALE
EA
P1.0
P1.1
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7

P0.0/A0
P0.1/A1
P0.2/A2
P0.3/A3
P0.4/A4
P0.5/A5
P0.6/A6
P0.7/A7
P2.0/A8
P2.1/A9
P2.2/A10
P2.3/A11
P2.4/A12
P2.5/A13
P2.6/A14
P2.7/A15
P3.0/RXD
P3.1/TXD
P3.2/INT0
P3.3/INT1
P3.4/T0
P3.5/T1
P3.6/WR
P3.7/RD

U2

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7D
8B
8C

ULN2803

AC

ANIMATING 00:00:03.050000 (CPU load 15%)

192411148 Exp 3 - Proteus 8 Professional - Source Code

File Project Build Edit Debug System Help

Schematic Capture Source Code

main.asm

```
1 Name K Hemant Kumar
2 Reg No 192411148
3 Code EGA1407
4 ORG 0000H
5 UP: MOV P2,#09H
6 ACALL DELAY
7 MOV P2,#0CH
8 ACALL DELAY
9 MOV P2,#09H
10 ACALL DELAY
11 MOV P2,#09H
12 ACALL DELAY
13 JUMP UP
14 DELAY: MOV R1,#10
15 H1: MOV R3,#255
16 H2: DJNZ R3,H1
17 DJNZ R1,H1
18 RET
19 END
```

VSM Studio Output

no errors
ASEMDDX.EXE Debug list
Processed 61 lines
Compiled successfully

Ready

Experiment 4 Output:

The image displays two screenshots from the Proteus 8 Professional software interface, showing the schematic capture and source code for an AT89C51 microcontroller project.

Schematic Capture: The top screenshot shows the schematic capture window. The AT89C51 microcontroller is connected to a 7-segment display. The connections are as follows:

- XTAL1 to P1.0
- XTAL2 to P1.1
- RST to P1.2
- PEER to P1.3
- EA to P1.4
- P1.5 to P1.6
- P1.7 to P1.7
- P2.0 to P2.0
- P2.1 to P2.1
- P2.2 to P2.2
- P2.3 to P2.3
- P2.4 to P2.4
- P2.5 to P2.5
- P2.6 to P2.6
- P2.7 to P2.7
- P3.0 to P3.0
- P3.1 to P3.1
- P3.2 to P3.2
- P3.3 to P3.3
- P3.4 to P3.4
- P3.5 to P3.5
- P3.6 to P3.6
- P3.7 to P3.7
- P4.0 to P4.0
- P4.1 to P4.1
- P4.2 to P4.2
- P4.3 to P4.3
- P4.4 to P4.4
- P4.5 to P4.5
- P4.6 to P4.6
- P4.7 to P4.7
- P5.0 to P5.0
- P5.1 to P5.1
- P5.2 to P5.2
- P5.3 to P5.3
- P5.4 to P5.4
- P5.5 to P5.5
- P5.6 to P5.6
- P5.7 to P5.7

Source Code: The bottom screenshot shows the source code window. The code is written in assembly language and is as follows:

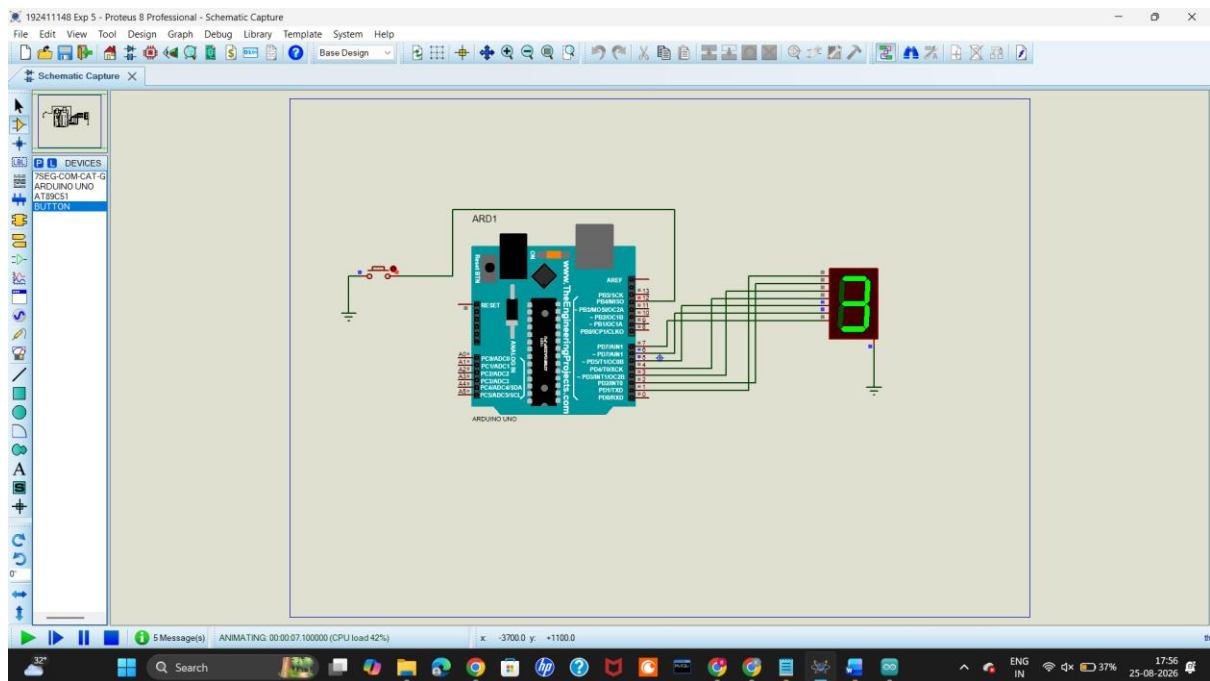
```
1 ;Name K Hemant Kumar
2 ;Reg No: 192411148
3 ;Code: ECA1407
4 ORG 000H
5 UP:MOV P2,#0C0H
6 ACALL DELAY
7 MOV P2,#0F0H
8 ACALL DELAY
9 MOV P2,#0A0H
10 ACALL DELAY
11 MOV P2,#0B0H
12 ACALL DELAY
13 MOV P2,#090H
14 ACALL DELAY
15 MOV P2,#0D0H
16 ACALL DELAY
17 MOV P2,#0E0H
18 ACALL DELAY
19 MOV P2,#0F0H
20 ACALL DELAY
21 MOV P2,#0D0H
22 ACALL DELAY
23 MOV P2,#0E0H
24 ACALL DELAY
25 DELAY:MOV R5,#10
26 H1:MOV R4,#10
27 H2:MOV R3,#255
28 H3:DJNZ R3,H3
29 DJNZ R4,H2
30 DJNZ R5,H1
31 RET
32 END
```

VSM Studio Output: The bottom screenshot shows the VSM Studio Output window. The output is as follows:

```
no errors
ASEMDDX.EVE Debug list
Processed 74 lines
Compiled successfully
```

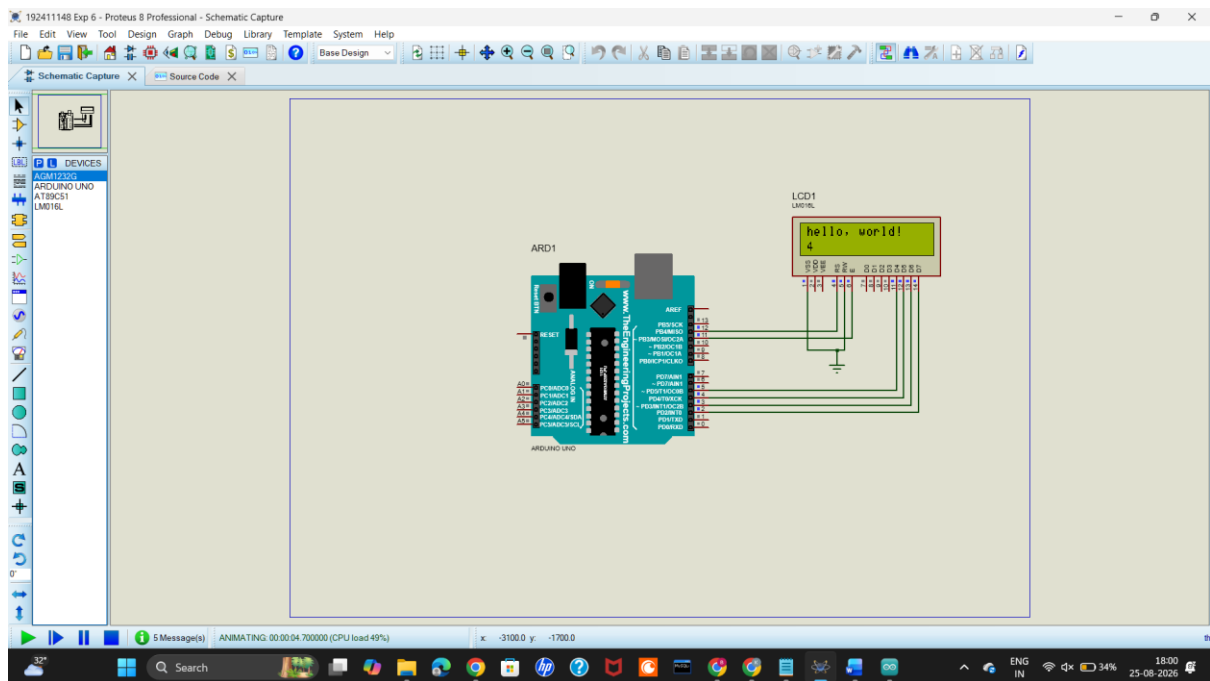
The VSM debugging session has ended.

Experiment 5 Output:



```
sketch_oct6a.ino
1 //The Coding://
2 //Name: K Hemanth Kumar
3 //reg no. 192411148
4 // CLASS: ECE
5
6 int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
7 int delay_time=200;
8
9 void setup() {
10
11 //configure pin2 as an input and enable the internal pull-up resistor
12 pinMode(12, INPUT_PULLUP);
13
14 pinMode(1,OUTPUT);
15 pinMode(2,OUTPUT);
16 pinMode(3,OUTPUT);
17 pinMode(4,OUTPUT);
18 pinMode(5,OUTPUT);
19 pinMode(6,OUTPUT);
20 pinMode(7,OUTPUT);
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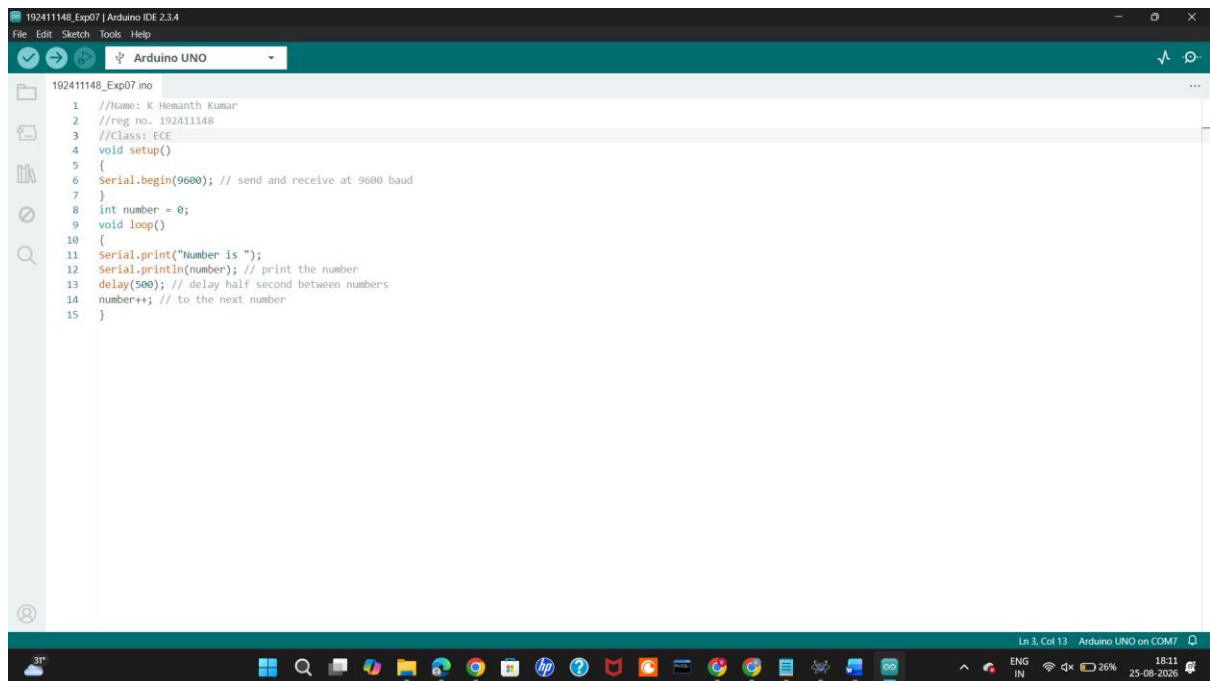
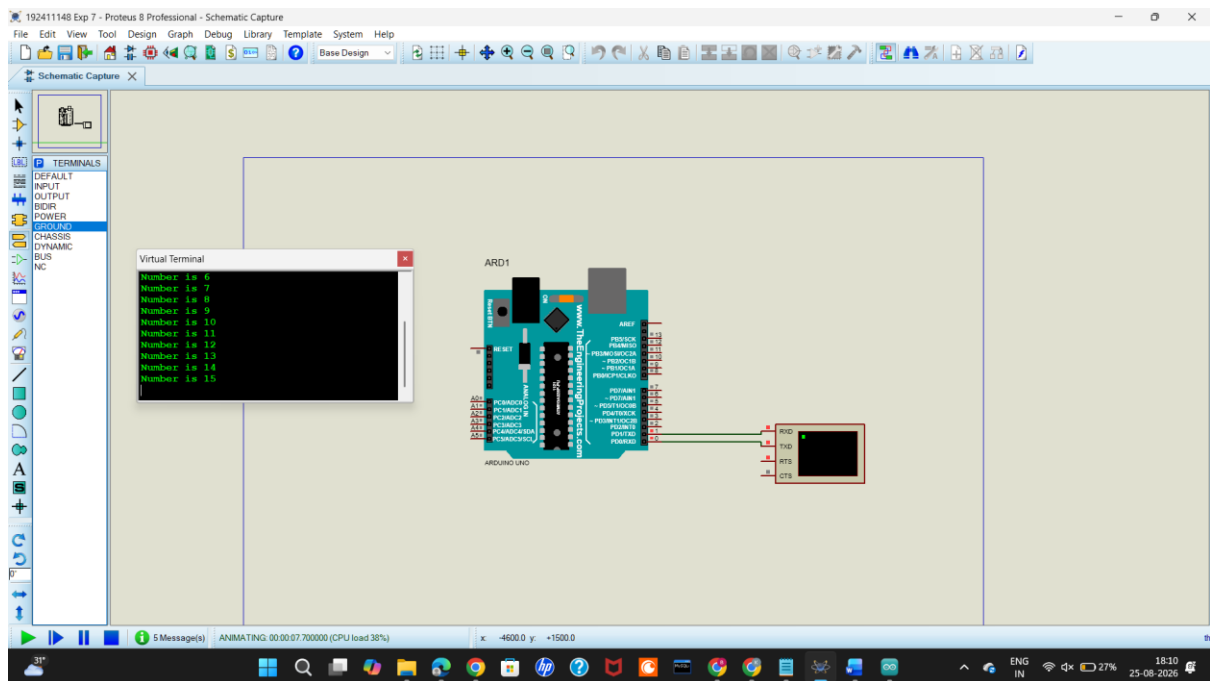
Experiment 6 Output:



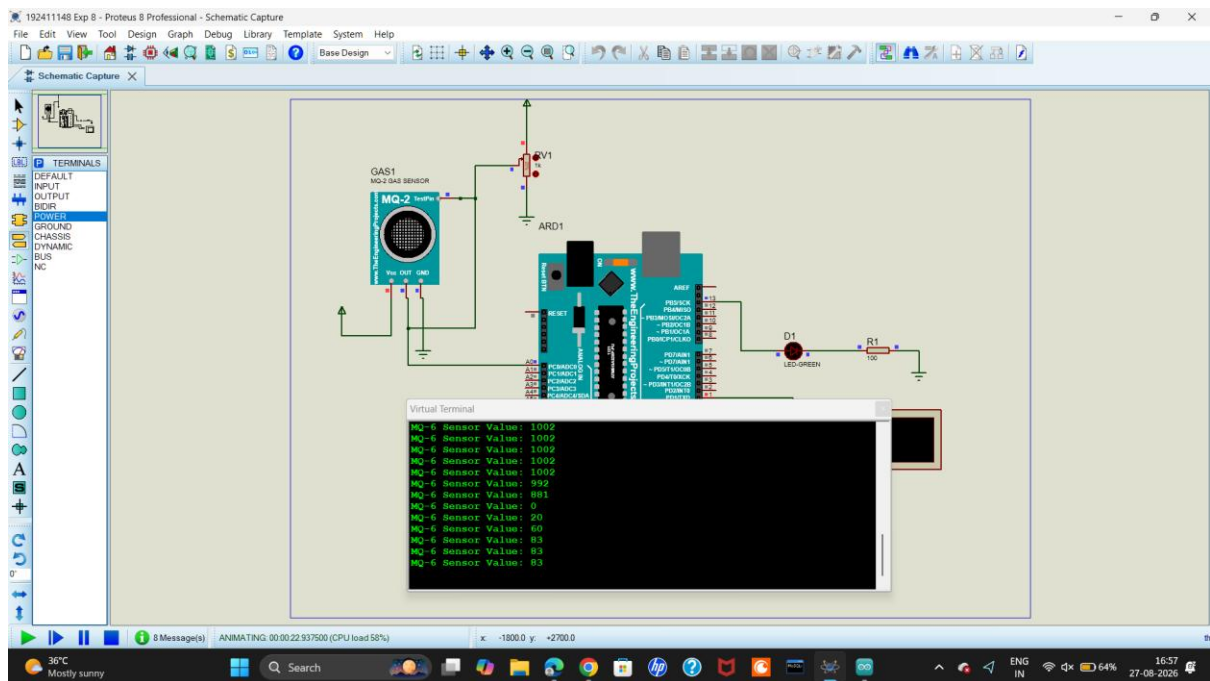
```
192411148_Exp06 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

192411148_Exp06.ino
1 //Name: K Hemanth Kumar
2 //reg no. 192411148
3 //Class: ECE
4
5 // include the library code:
6 #include <LiquidCrystal.h>
7
8 // initialize the library by associating any needed LCD interface pin
9 // with the arduino pin number it is connected to
10 const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
11 LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
12
13 void setup() {
14   // set up the LCD's number of columns and rows:
15   lcd.begin(16, 2);
16   // Print a message to the LCD.
17   lcd.print("hello, world!");
18 }
19
20 void loop() {
21   // set the cursor to column 0, line 1
22   // (note: line 1 is the second row, since counting begins with 0):
23   lcd.setCursor(0, 1);
24   // print the number of seconds since reset:
25   lcd.print(millis() / 1000);
26 }
27
28
```

Experiment 7 Output:



Experiment 8 Output:



```
192411148_Exp08 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

192411148_Exp08.ino
1 //Name: K Hemanth Kumar
2 //reg no. 192411148
3 //Class: ECE
4 #define GreenLed 13
5 #define Sensor A0
6
7 void setup()
8 {
9   pinMode(GreenLed, OUTPUT);
10  pinMode(Sensor, INPUT);
11
12  Serial.begin(9600);
13 }
14
15 void loop()
16 {
17   int value = analogRead(Sensor);
18
19   Serial.print("MQ-2 Sensor Value: ");
20   Serial.println(value);
21
22   if (value > 600)
23   {
24     digitalWrite(GreenLed, HIGH);
25   }
26   else
27   {
28     digitalWrite(GreenLed, LOW);
29   }
30
31   delay(500);
32 }
```

Updates are available for some of your boards. [LATER] [INSTALL MANUALLY] [INSTALL ALL]

[illegible]

```

1 //Name: K Hemanth Kumar
2 //reg no. 192411148
3 //Class: ECE
4 int trig = 10;
5 int echo = 9;
6
7 long duration;
8 int cm;
9
10 void setup()
11 {
12     pinMode(trig, OUTPUT);
13     pinMode(echo, INPUT);
14
15     Serial.begin(9600);
16 }
17
18 void loop()
19 {
20     // Send trigger pulse
21     digitalWrite(trig, LOW);
22     delayMicroseconds(2);
23
24     digitalWrite(trig, HIGH);
25     delayMicroseconds(10);
26
27     digitalWrite(trig, LOW);

```

Output

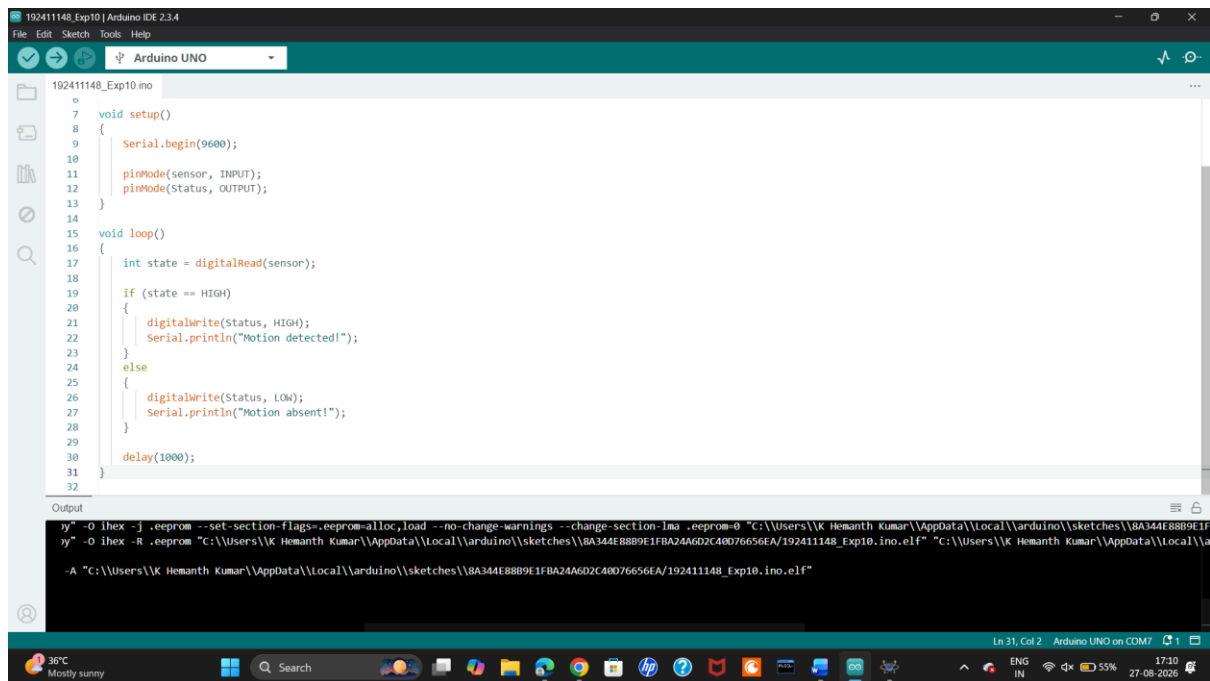
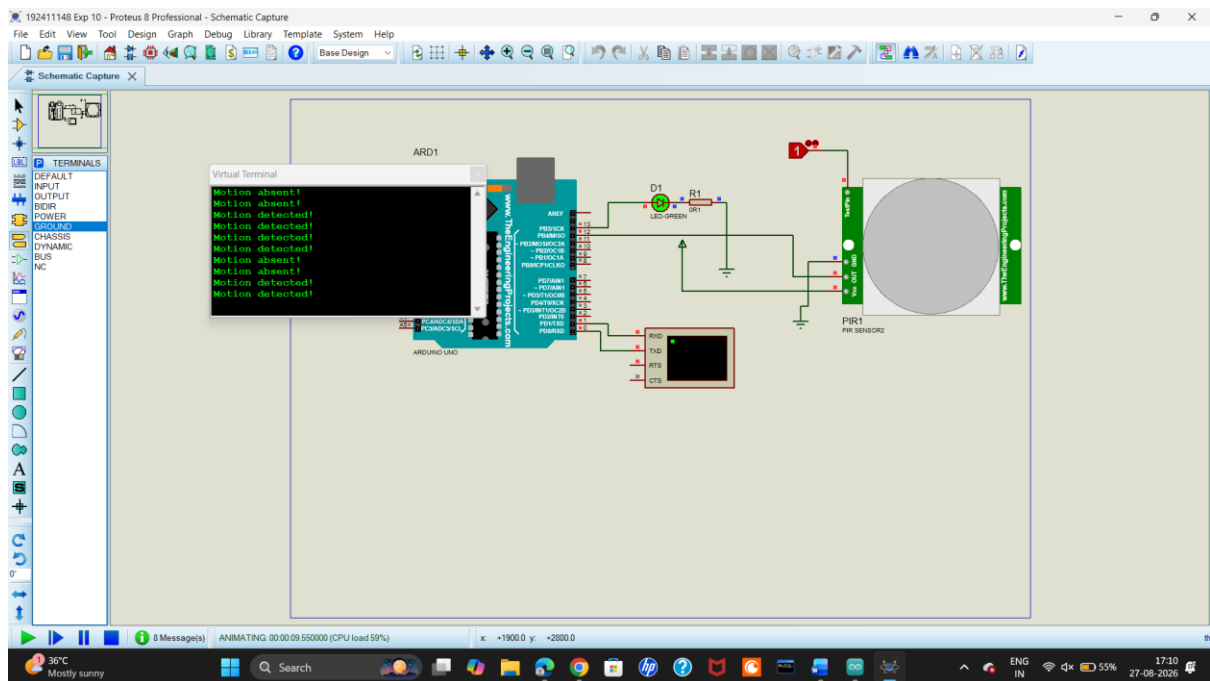
```

.0-atmel3.6.1-arduino/bin/avr-objcopy -O ihex -j .eeprom --set-section-flags=.eeprom=alloc,load --no-change-warnings --change-section-lma .eeprom=0 "C:\\Users\\K Hemanth Kumar\\AppData\\Local\\Arduino15\\sketches\\192411148_Exp09.ino.elf" "C:\\Users\\K Hemanth Kumar\\AppData\\Local\\Arduino15\\sketches\\192411148_Exp09.ino.elf"
.0-atmel3.6.1-arduino/bin/avr-objcopy -O ihex -R .eeprom "C:\\Users\\K Hemanth Kumar\\AppData\\Local\\arduino\\sketches\\081847222CC1DA1F297A074A79548E51\\192411148_Exp09.ino.elf" "C:\\Users\\K Hemanth Kumar\\AppData\\Local\\arduino\\sketches\\081847222CC1DA1F297A074A79548E51\\192411148_Exp09.ino.elf"
.0-atmel3.6.1-arduino/bin/avr-size -A "C:\\Users\\K Hemanth Kumar\\AppData\\Local\\arduino\\sketches\\081847222CC1DA1F297A074A79548E51\\192411148_Exp09.ino.elf"
.. Maximum is 2048 bytes.

```

Ln 1, Col 1 Arduino UNO on COM7

Experiment 10 Output:



Experiment 11 Output:

The screenshot displays the uVision IDE interface during the execution of a C program for an 8051 microcontroller. The main window shows the source code for 'main.c' with the following content:

```
1 //Name:K Hemanth Kumar
2 //seg no. 192411148
3 //Class: ECE
4 #include <reg51.h>
5
6 sbit LED1 = P1^0;
7 sbit LED2 = P1^1;
8
9 void delay()
10 {
11     unsigned int i,j;
12     for(i=0;i<200;i++)
13         for(j=0;j<1000;j++);
14 }
15
16
17 void main()
18 {
19     while(1)
20     {
21         LED1 = 1;
22         LED2 = 0;
23         delay();
24     }
25 }
```

The 'Registers' window on the left shows the state of the 8051 registers, with the PC register at 0x000E and the SP register at 0x0009. The 'Parallel Port 1' and 'Parallel Port 0' windows show the current port values: P1 is 0xFD and P0 is 0xFF. The 'Command' window at the bottom shows the execution progress, including the command 'Load *C:\Users\K Hemanth Kumar\Downloads\embedded\EXP 11 F\Objects\exp 11*' and the simulation status 'Simulation'.

Experiment 12 Output:

The screenshot displays the uVision IDE interface during the execution of a C program. The main window shows the source code in `main.c`:

```
1 // NAME: K Hemant Kumar
2 // ROLL NO: 192411149
3 // CLASS: ECE
4 #include <reg51.h>
5
6 void main()
7 {
8     unsigned int num1, num2;
9     unsigned long int product;
10
11     while(1)
12     {
13         // Pot 2
14         long int) num1 * num2;
15
16         // Pot 3
17         // Pot 1
18         // Pot 0
19         // Pot 2
20         P2 = (product >> 16) & 0xFF;
21         P3 = (product >> 24) & 0xFF;
22     }
23 }
```

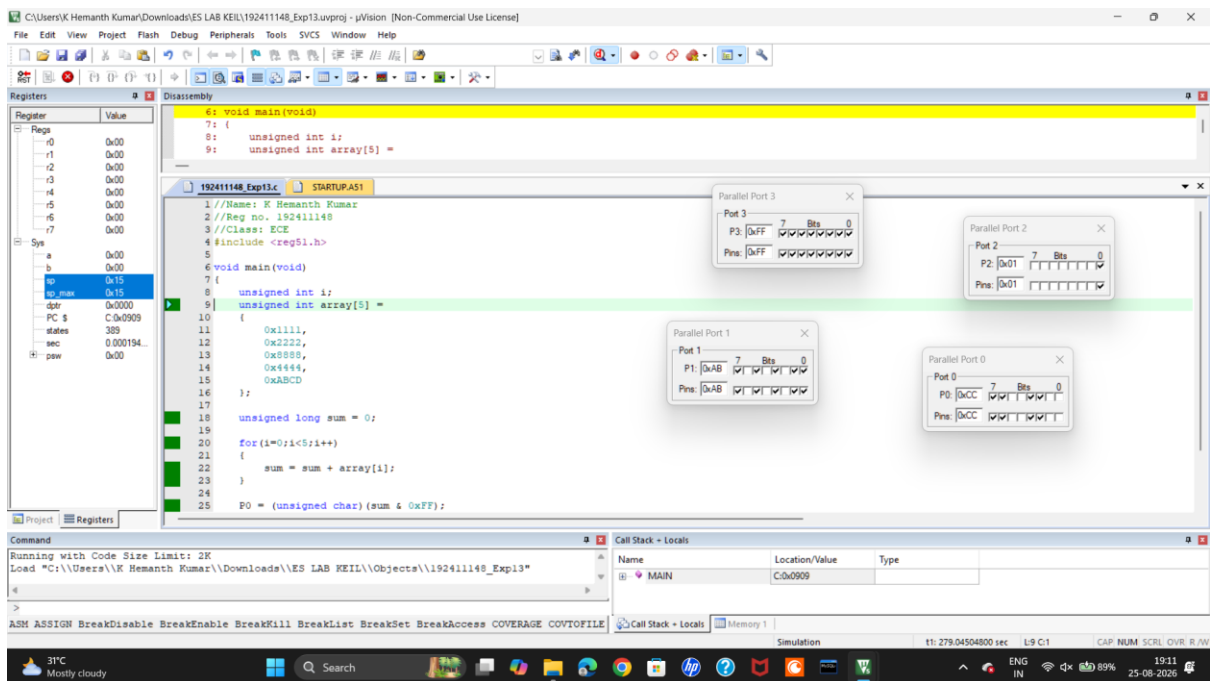
Four floating windows show the state of Parallel Ports 0, 1, 2, and 3. Each window displays the port number, the value of the port, and the state of the pins (P0, P1, P2, P3).

The bottom panel shows the Command window with the load command:

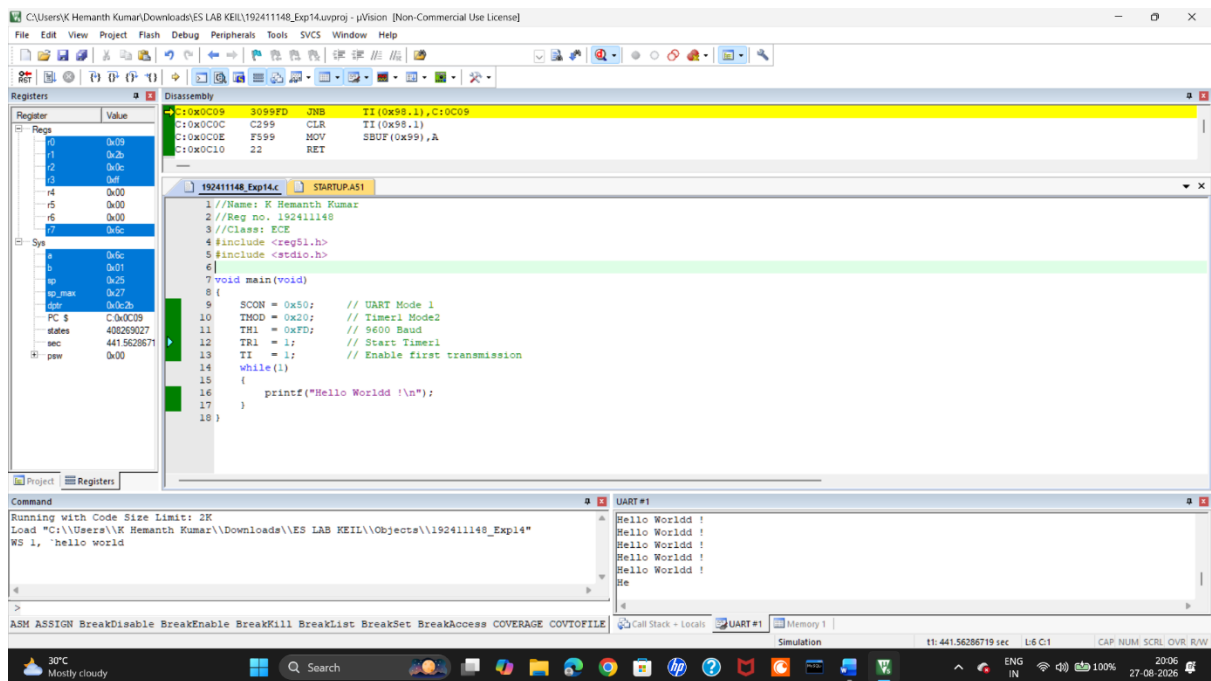
```
Running with Code Size Limit: 2K
Load "C:\\Users\\K Hemant Kumar\\Downloads\\embedded\\exp 12\\Objects\\exp 12"
>
```

The Call Stack window shows the current function `MAIN` at location `C:\\Users\\K Hemant Kumar\\Downloads\\embedded\\exp 12\\Objects\\exp 12`.

Experiment 13 Output:



Experiment 14 Output:



Experiment 15 Output:

The screenshot displays the Keil uVision IDE interface for a project named "192411148_Exp15.c". The main window shows the disassembly of the C code. The code includes a delay function and a main function that initializes the UART module and prints "Hello World". The output window shows the execution results, including the printed text and the interrupt count.

Registers:

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
sp	0x00
sp_max	0x00
dptr	0x0000
PC	0x00CA7
status	339
sec	0.000194
pew	0x00

Disassembly:

```
19: void main(void)
20: {
21:   SCON = 0x50; // UART mode 1, 8-bit, REN enabled
22:   TH1 = 0xFD; // Baud rate 9600
23:   TR1 = 1; // Start Timer1
24:   IE = 0x80; // EA = 1, ES = 1 (enable serial interrupt)
25:   TI = 1; // Enable printf
26:   while(1)
27:   {
28:     printf("Hello World\n");
29:     delay();
30:   }
31: }
32: }
33: }
```

Command:

```
Running with Code Size Limit: 2K
Load "C:\Users\K Remanth Kumar\Downloads\ES LAB KEIL\Exp15\Objects\192411148_Exp15"
```

UART #1:

```
Hello World
Hello World
>>> INTERRUPT RECEIVED: 3
Hello World
```

Simulation:

11:720.77048000 sec L21 C1 CAP NUM SCBL OVR R/W

Experiment 16 Output:

The screenshot displays the uVision IDE interface with the following components:

- Registers:** A list of registers (r0-r7, sp, sp_max, dptr, PC, states, sec, psw) and their values.
- Disassembly:** A list of instructions with addresses and values. The highlighted instruction is `133: 80FE SJMP C:0014`.
- Source Code:** A C program titled "192411148 Exp16.c" with the following code:

```
1 //Name: R Remanath Kumar
2 //Reg No. 192411148
3 //Class: ECE
4 #include <reg51.h>
5 void main(void)
6 {
7     unsigned char hexa = 0x08;
8     unsigned char hundrede, tens, units;
9     ACC = hexa;
10    B = 10;
11    ACC = ACC / B;
12    units = B;
13    B = 10;
14    ACC = ACC / B;
15    tens = B;
16    hundreds = ACC;
17    P0 = units;
18    P1 = tens;
19    P2 = hundreds;
20    while(1);
21 }
```
- Command:** A list of commands. The highlighted command is `Load "C:\\Users\\K Remanath Kumar\\Downloads\\ES LAB KEIL\\Objects\\Exp 16"`.
- Call Stack - Locals:** A table showing the current function and its location/value.

Name	Location/Value	Type
MAIN	C:00800	
- Status Bar:** Shows simulation time (11:2055.66550250 sec), L3 C13, and temperature (30°C).

Experiment 17 Output:

The image displays two screenshots of the uVision IDE, showing the execution of an AVR program. The top screenshot shows the initial setup, and the bottom screenshot shows the program running.

Top Screenshot:

- Registers:** Shows the initial state of registers. The PC register is at 0x0000, and the SP register is at 0x0000. The program counter (PC) is at 0x0000.
- Disassembly:** Shows the assembly code for the program. The code includes a main function that initializes the serial port and enters a loop.
- Interrupt System:** Shows the interrupt configuration. The interrupt source is set to P3.2/INT0, and the interrupt mode is set to Edge Triggered (IT0).
- Parallel Port 1:** Shows the parallel port configuration. The port is set to P1, and the pins are configured for output.
- Serial Channel:** Shows the serial port configuration. The mode is set to 8-Bit Shift Register, and the baud rate is set to 2000000.

Bottom Screenshot:

- Registers:** Shows the state of registers after execution. The PC register is at 0x000F, and the SP register is at 0x0000. The program counter (PC) is at 0x000F.
- Disassembly:** Shows the assembly code for the program. The code includes a main function that initializes the serial port and enters a loop.
- Interrupt System:** Shows the interrupt configuration. The interrupt source is set to P3.2/INT0, and the interrupt mode is set to Edge Triggered (IT0).
- Parallel Port 1:** Shows the parallel port configuration. The port is set to P1, and the pins are configured for output.
- Serial Channel:** Shows the serial port configuration. The mode is set to 8-Bit Shift Register, and the baud rate is set to 2000000.

Experiment 18 Output:

Experiment 24 Output:

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL `edaplayground.com/v/epzt`. The interface includes a top navigation bar with buttons for 'New', 'Run', 'Save', and 'Copy', along with links to 'Resources', 'Community', 'Help', 'Playgrounds', and 'Profile'. On the left, a sidebar lists 'Languages & Libraries' and 'Tools & Simulators', with a section for 'Examples' including 'using EDA Playground', 'VHDL', 'Verilog/SystemVerilog', 'UVM', 'EasierUVM', 'SVAUnit', 'SVUnit', 'VUnit (Verilog/SV)', 'VUnit (VHDL)', 'TL-Verilog', 'e + Verilog', 'Python + Verilog', 'Python + VHDL', 'Python Only', and 'C++/SystemC'. The main workspace is divided into two panels: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' panel contains the following C++ code:

```
1 #include <systemc.h>
2 SC_MODULE(ClockExample) {
3   SC_In<bool> clk;
4   void process() {
5     while (true) {
6       wait();
7       std::cout << "Time: " << sc_time_stamp()
8         << " - Clock Triggered" << std::endl;
9     }
10  }
11  SC_CTOR(ClockExample) {
12    SC_THREAD(process);
13    sensitive << clk.pos();
14  }
15 };
16 int sc_main(int argc, char* argv[]) {
17   sc_clock clock("clock", 1, SC_NS);
18   ClockExample module("module");
19   module.clk(clock);
20   sc_start(5, SC_NS);
21   return 0;
22 }
```

The 'design.cpp' panel contains the following C++ code:

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

Below the code panels is a 'Log' section showing the execution output. The output starts with a timestamp and the command used to run the simulation:

```
[2026-08-28 07:58:00 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
```

Followed by the compilation status:

```
Compile done. Starting run...
```

Then, the SystemC version and copyright information are displayed:

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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```

The main output shows the clock triggering at regular intervals:

```
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
```

The output ends with the word 'Done'.

The bottom of the screenshot shows a Windows taskbar with the system clock at 13:24 on 28-08-2026, and the battery level at 56%.

Experiment 25 Output:

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL `edaplayground.com/x/ZxzQ`. The interface includes a sidebar on the left with a menu for "Examples" containing various simulation and development options like VHDL, Verilog, UVM, and SystemC. The main workspace is divided into two code editors: "testbench.cpp" and "design.cpp". The "testbench.cpp" editor contains a C++ code snippet for a SystemC-based producer-consumer simulation. The "design.cpp" editor is currently empty, showing only placeholder comments. Below the code editors is a "Log" window displaying the output of the simulation, which shows a sequence of "Produced" and "Consumed" messages. The bottom of the image shows a Windows taskbar with the system clock indicating 13:27 on 28-08-2026.

```
//Name: K Hemanth Kumar
//Reg no. 192411148
//Class: ECE
#include <systemc.h>
// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;
    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }
    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};
// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;
    void consume() {
        for (int i = 1; i <= 5; i++) {
            int val = in.read();
            cout << "Consumed: " << val << endl;
        }
    }
    SC_CTOR(Consumer) {
        SC_THREAD(consume);
    }
};
int main() {
    Producer p;
    Consumer c;
    p.out <-> c.in;
    p.start();
    c.start();
    wait(10, SC_SEC);
    return 0;
}
```

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Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5

Experiment 26 Output:

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL `edaplayground.com/v/kkz8`. The interface includes a top navigation bar with buttons for 'New', 'Run', 'Save', and 'Copy', along with links to 'Resources', 'Community', 'Help', 'Playgrounds', and 'Profile'. On the left, a sidebar lists 'Languages & Libraries' and 'Tools & Simulators', with a section for 'Examples' including various languages like VHDL, Verilog, and Python. The main workspace is divided into two panes: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' pane contains C++ code for a SystemC-based testbench, including headers, namespace declarations, and a main function that sets up a target and a transport channel. The 'design.cpp' pane is currently empty, showing only placeholder comments. Below the code panes, a 'Log' tab is active, displaying the output of the simulation. The log shows the compilation command, the SystemC version (2.3.3-Accellera), and the simulation results, which indicate that the target received the value 555. The bottom of the image shows a Windows taskbar with the date and time set to 13:28 on 28-08-2026.

```
//Name: K Hemanth Kumar
//Reg no. 192411148
//Class: ECE
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>
using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module {
    tlm_utils::simple_target_socket<Target> socket;
    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this, &Target::b_transport);
    }
    void b_transport(tlm_generic_payload& trans, sc_time& delay) {
        int* data = (int*) trans.get_data_ptr();
        cout << "Target received: " << *data << endl;
        delay += sc_time(10, SC_NS);
    }
};
```

```
[2026-08-28 08:01:42 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

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Initiator sending: 555
Target received: 555
Done
```

Experiment 27 Output:

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL `edaplayground.com/v/QEyT`. The interface includes a top navigation bar with buttons for 'New', 'Run', 'Save', and 'Copy', along with links to 'Resources', 'Community', 'Help', 'Playgrounds', and 'Profile'. On the left, a sidebar lists 'Languages & Libraries' and 'Tools & Simulators', with a section for 'Examples' showing various simulation environments like VHDL, Verilog/SystemVerilog, UVM, and others. The main workspace is divided into two code editors: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' editor contains C++ code for a SystemC testbench, including headers, a semaphore definition, and a task function. The 'design.cpp' editor is currently empty. Below the code editors, a terminal window shows the output of the compilation and simulation process. The terminal output includes the compilation command, the SystemC version (2.3.3-Accellera), copyright information, and the simulation results showing 'Task1 Entered Critical Section' at 1s, 4s, and 7s. The terminal also shows a 'Done' status at the bottom.

```
//Name: K Hemanth Kumar
//Reg no: 192411148
//Class: ECE
#include <systemc.h>
SC_MODULE(binary_semaphore) {
    int sem;
    sc_event ev;
    void task1() {
        while(true) {
            wait(1, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }
};
```

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

```
[2026-08-28 08:01:59 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

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Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
Done
```

Experiment 28 Output:

The screenshot displays the EDA Playground web interface. The top navigation bar includes 'New', 'Run', 'Save', and 'Copy' buttons. The left sidebar lists various examples and languages, including 'using EDA Playground', 'VHDL', 'Verilog/SystemVerilog', 'UVM', 'EasierUVM', 'SVAUnit', 'SVUnit', 'VUnit (Verilog/SV)', 'VUnit (VHDL)', 'TL-Verilog', 'e + Verilog', 'Python + Verilog', 'Python + VHDL', 'Python Only', and 'C++/SystemC'. The main editor area shows two files: 'testBench.cpp' and 'design.cpp'. The 'testBench.cpp' file contains the following code:

```
1 //Name: K Hemanth Kumar
2 //Reg no. 192411148
3 //Class: ECE
4 #include <systemc.h>
5 SC_MODULE(counting_semaphore) {
6     int sem;
7     void task() {
8         while(true) {
9             wait(1, SC_SEC);
10            if(sem > 0) {
11                sem--;
12                cout << "Task Accessing Resource at "
13                    << sc_time_stamp()
14                    << " Remaining: " << sem << endl;
15                wait(2, SC_SEC);
16                sem++;
17            }
18        }
19    }
20    SC_CTOR(counting_semaphore) {
21        sem = 2;
22    }
23 }
```

The 'design.cpp' file contains the following code:

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

The bottom log window shows the output of the simulation:

```
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Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
Done
```

The bottom status bar shows the system temperature as 35°C, the weather as Sunny, and the time as 13:28 on 28-08-2026.

Experiment 29 Output:

The screenshot displays the EDA Playground web interface. The top navigation bar includes the EDA Playground logo, a user profile, and links for New, Run, Save, and Copy. The left sidebar contains a menu with 'Languages & Libraries', 'Tools & Simulators', and 'Examples'. The main area is divided into two code editors: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' editor contains C++ code for a SystemC module named 'ECE' with two interrupt handlers, 'low_ISR' and 'high_ISR'. The 'design.cpp' editor is empty. Below the code editors is a terminal window showing the output of the compilation and simulation. The terminal output indicates that the code was compiled successfully and the simulation ran for 2 seconds, with the 'Done' status highlighted in blue. The bottom status bar shows the system temperature as 35°C and the date as 28-08-2026.

```
//Name: K Hemanth Kumar
//Reg no. 192411148
//Class: ECE
#include <system.h>
SC_MODULE(priority_interrupt) {
    sc_event low_int, high_int;
    void low_ISR() {
        while(true) {
            cout << "Low Priority Interrupt at "
            << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }
    void high_ISR() {
        while(true) {
            cout << "High Priority Interrupt at "
            << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }
}
```

```
// Code your design here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
```

Log Share

```
[2026-08-28 08:02:38 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

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Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
Done
```

35°C Sunny 13:28 28-08-2026

Experiment 30 Output:

The screenshot displays the EDA Playground web interface. The left sidebar contains navigation links for Languages & Libraries, Tools & Simulators, and Examples. The main editor area is split into two panes: 'testBench.cpp' and 'design.cpp'. The 'testBench.cpp' pane contains C++ code for two tasks, task1 and task2, which are scheduled using SC_THREAD and SC_THREAD macros. The 'design.cpp' pane contains a placeholder for the user's design code. Below the code panes, the 'Log' tab shows the execution output, which displays the timestamps for each task's execution. The output shows that task1 runs at 0s, 1s, 2s, 3s, 4s, and 5s, while task2 runs at 1s, 2s, 3s, 4s, and 5s. The 'Done' button is visible at the bottom of the log.

```
//Name: K Hemanth Kumar
//Reg no. 192411148
//Class: ECE
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
```

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

Task1 Running at 0 s
Task2 Running at 1 s
Task1 Running at 1 s
Task2 Running at 2 s
Task1 Running at 2 s
Task2 Running at 3 s
Task1 Running at 3 s
Task2 Running at 4 s
Task1 Running at 4 s
Task2 Running at 5 s
Task1 Running at 5 s
Task2 Running at 5 s

Done